Al-Balqa Applied University

جامعة البلقاء التطبيقية

Midterm Exam-C

Course Title	General Physics II	Course Code	30201102
Semester	2st	Academic Year	2024-2025
Exam Date	14/4/2025	Day Wednesday	Time 14:30-15:20
Student Name	[redacted]	Student ID No. [redacted]	ction monday wednesday 11:30-7:00

GENERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:** _____

Number of Questions	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
12	Multiple Choices Questions	CLIO 1, 2, 7, 12	24(100%)	22		
3	Computational Questions	CLIO 1, 2, 7, 12	6(100%)	6		
Total			30(100%)			28 /30

Moderation Review			
Comments			
Marks Reviewed by		Signature & Date	

Part one: Put the correct choice (A, B, C, or D) for each question in this table:

Question no.	1	2	3	4	5	6	7	8
Answer	A	A	d	d	d	C	C	d
Question no.	9	10	11	12	MCQ /24	Q13,14 &15/6	Total/30	
Answer	b	b	d	C				

constants

$q_e = -1.6 \times 10^{-19} C, q_p = +1.6 \times 10^{-19} C, m_e = 9.11 \times 10^{-31} kg, m_p = 1.672 \times 10^{-27} kg,$

$\epsilon_o = 8.85 \times \frac{10^{-12} C^2}{N \cdot m^2}, k_e = 8.987 \times \frac{10^9 N \cdot m^2}{C^2}$

- يتكون الجزء الأول من 12 سؤال إختيار من متعدد، بعد اختيار رمز الإجابة الصحيحة يتم تثبيتها في الجدول المخصص بالصفحة الأولى

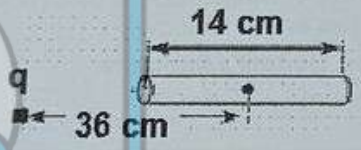
Q.1) A ring of radius (a) carries a uniformly distributed positive total charge (Q). The electric field due to the ring at a point P lying a distance (x) from its center along the central axis perpendicular to the plane of the ring is

$$(E_P = \frac{k_e x}{(a^2+x^2)^{\frac{3}{2}}} Q).$$

The electric field as $(x << a)$ along the x axis is approximately equal to

a. $E_x \approx \frac{k_e Q}{a^3} x$ b. $E_x \approx \frac{k_e Q}{x^3} a$ c. $E_x \approx \frac{k_e Q}{x^2}$ d. $E_x \approx k_e Q\, x$

Q.2) A rod **14.0 cm** long is uniformly charged and has a total charge of **-22.0 μC**. Determine the magnitude of the electric force exerted on the point charge **(q = −17 μC)** located along the axis of the rod at a point **36.0 cm** from its center as shown in the figure below.



a. 27 N b. 1.6 N c. 40 N d. 54 N

Q.3) Two point charges are placed along a horizontal axis with the following values and positions: **+3.0 μC, at x = 0 cm and -7.0 μC, at x = 20 cm**. At what point along the x axis is the electric field **zero**?

a. 8.0 cm b. -44 cm c. -24 cm d. -38 cm

Q.4) Two electrons are separated by **one cm**. What is the ratio of the electric force to the gravitational force between them?

$(the\ gravitational\ constant\ G = 6.67 \times 10^{-11}\ N \cdot m^2/kg^2)$

a. 2.3×10^2 b. 1.2×10^{36} c. 3.1×10^{22} d. 4.1×10^{42}

Q.5) An airplane is flying through a thundercloud at a height of **2000 m**. If there is a charge concentration of **+40 C at height 3000 m** within the cloud and **-40 C at height 1000 m**, what is the electric field in **(N/C)** at the aircraft?

a. 718 960 (up) b. 539 400 (up) c. 539 400 (down) d. 718 960 (down)

Q.6) Electrons in a particle beam each have a kinetic energy of 3.2×10^{-17} J. What is the magnitude of the electric field that will stop these electrons in a distance of 0.1 m?

a. 1375 N/C b. 1 000 N/C c. 2 000 N/C d. 4 000 N/C

Q.7) The electric field in a cathode ray tube is supposed to accelerate electrons from 0 to 1.60×10^{7} m/s in a distance of 2.00 cm. What electric field is required?

a. 96 224 N/C b. 18 200 N/C c. 36 400 N/C d. 72 800 N/C

Q.8) Q_1 has 50 electric field lines radiating outward and Q_2 has 100 field lines converging inward. What is the ratio $\frac{Q_1}{Q_2}$?

a. 2 b. -2 c. 1/2 d. -1/2

Q.9) Two identical spheres one carry a charge of $(-40\ \mu c)$, and the another also carry a charge of $(-40\ \mu c)$ The spheres are separated by a distance of $(1\ m)$. What is the electric force between the spheres?

a. 28.8 N (repulsive) b. 14.4 N (repulsive) c. 28.8 N (attractive) d. 14.4 N (attractive)

Q.10) A ping-pong ball covered with a conducting graphite coating has a mass of $(5\ gram)$ and a charge of $(4\ \mu c)$. What electric field directed upward will exactly balance the weight of the ball? $(g = 9.8\ m/s^2)$.

a. $8.22 \times 10^2 N/C$ b. $1.225 \times 10^4 N/C$ c. $2.11 \times 10^{-7} N/C$ d. $3.6 \times 10^4 N/C$

Q.11) A spherical volume of space has an electric field of intensity 100 N/C directed radially outward from it surface of radius 0.600 m. What is the net charge enclosed within this surface?

a. 6.8 nC b. 4.0 nC c. -6.8 nC d. -4.0 nC

Q.12) Three equal positive charges (+q, +q, and +q) are placed on the x-axis, at x = 0, x = 2 m, and x = 4 m, respectively. Of the following points, which has the greatest magnitude electric field?

a. x = 1m b. x = 3m c. x = 5m d. The same magnitude at all positions.

- يتكون الجزء الثاني من 3 أسئلة حل، يتم حل السؤال مع الخطوات في المكان المخصص للإجابة عند السؤال في هذه الورقة.

Q.13) If a charge $(Q = \pi\ \mu C)$ is uniformly distributed throughout a sphere with radius $(r = 1\ cm)$, the volume charge density (ρ) is

$Q = \pi \times 10^{-6} C$
$r = 0.01 m$
$\rho = ?$

$\rho = \frac{Q}{V} = \frac{Q}{4/3 \pi r^3} = \frac{3Q}{4\pi r^3}$

$\rho = \frac{3\pi \times 10^{-6}}{4\pi (0.01)^3}$

$\rho = 0.75\ C/m^3$

Q.14) a charged particle has a charge $(q = 0.1\ mC)$, and mass $(m = 0.02\ kg)$ is inserted in a uniform electric field $(E = 10 \times 10^3\ C)$, calculate the acceleration of a charged particle.

$q = 1 \times 10^{-3} C$
$m = 0.02 kg$
$E = 10 \times 10^3\ N/C$
$a = ?$

$\Sigma F = Fe = ma$
$qE = ma$
$a = \frac{qE}{m} = \frac{(1 \times 10^{-3})(10 \times 10^3)}{(0.02)}$
$a = 500\ m/s$

Q.15) Consider a long, cylindrical charge distribution of radius $(R = 10\ cm)$, with a uniform charge density $(\rho = 8.85 \times 10^{-12}\ C/m^3)$. Find the electric field at distance $(r = 6\ cm)$ from the axis.

$R = 0.1 m$
$\rho = 8.85 \times 10^{-12}\ C/m^3$
Eat $r = 0.06 m$

$E = \frac{\rho r}{2\epsilon_0}$

$E = \frac{(8.85 \times 10^{-12})(0.06)}{2(8.85 \times 10^{-12})} = \frac{r}{2}$

$E = 0.03\ N/C$